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RESEARCH

# Thermoeconomics Meets Business Science: Systemic Exergy Management (SYMΞX) as a New Theoretical and Flexible Framework for Sustainability

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***Abstract*** *Historically, corporate assessment of performance has placed a significant emphasis on financial measures, while largely ignoring the broader environmental and resource efficiency implications of organizational operations. This paper introduces a novel framework that integrates thermoeconomic principles into performance measurement. Systemic Exergy Management (SYMΞX) The study employs a theoretical construction approach, deriving conceptual claims from an extensive literature review to bridge the gap between thermoeconomics and existing frameworks. These claims are then synthesized into a novel theoretical construct, designated the SYMΞX framework, through the process of abductive reasoning. This flexible paradigm focuses on measuring the effectiveness of resource consumption in organizations using exergy analysis. Moreover, it enables the development of sustainability performance indicators for assessing impacts on the environment, economy, society, and technology. SYMΞX offers a more comprehensive perspective on organizational performance, accounting not only for financial outcomes but also for resource efficiency and environmental impacts. The study posits that SYMΞX has the potential to advance business science in two main ways: by fostering ethical corporate practices and by enhancing performance measurement. In conclusion, the framework promotes systemic sustainability while offering new avenues for empirical validation and industry-specific applications, thereby ensuring adaptability to evolving business environments.*

**Keywords** Exergy analysis · Organizational flexibility · Organizational performance measurement · Resource efficiency · Sustainability performance indicators · Systemic exergy management (SYMΞX) · Thermoeconomics

**JEL Classification** D81 · L61 · L21 · L25 · D24 · Q01

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## Introduction

### Context and Motivation

The persistently mounting challenges of resource scarcity and climate change necessitate a profound transformation in how companies address sustainability. This transformation must encompass a recognition of the multifaceted and multidimensional nature of sustainability, encompassing environmental, social, economic, and technological dimensions. Despite their utility in understanding market dynamics, conventional economic models are often inadequate for addressing environmental concerns. Such



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constraints are a consequence of a perspective that prioritizes rapid financial gain, frequently at the expense of long-term environmental degradation (Dey et al., 2020). The prevailing approach to modeling resources such as labor and materials is to treat them as mere economic inputs, with a primary focus on monetary transactions and financial exchanges. This approach fails to consider the physical realities of resource utilization, including the environmental consequences of resource extraction and processing, as well as the depletion of finite natural resources. The discounting strategies employed in traditional models frequently prioritize immediate earnings over long-term consequences. Such an approach may result in unsustainable behaviors that prioritize immediate financial gain over the long-term environmental impact (Brozovic, 2020). The prevailing practices and standards in the field of sustainability are unable to effectively reflect the evolving nature of sustainability challenges. By failing to account for the evolving nature of environmental challenges and resource limitations, these practices offer a static representation of an environmental baseline that companies are expected to adhere to (Lakner et al., 2024). Conventional models hardly give the environment's ecological carrying capacity any thought. This can cause companies to surpass the capacity of ecosystems to absorb pollutants and trash, therefore degrading the surroundings (Deng et al., 2024). These restrictions highlight the necessity for the development of a novel economic paradigm that incorporates environmental concerns into the decision-making processes of corporations. This is where thermoeconomics, a field that integrates economic principles with the laws of thermodynamics, presents a promising avenue for resolution. The field of thermoeconomics provides a framework that recognizes the physical limits that control resource consumption and environmental impact. This framework combines the ideas of economics with the laws of thermodynamics (Jamil et al., 2020).

Exergy, a measure of the useful energy available in a system, is a fundamental concept in thermoeconomics (Querol et al., 2013). In contrast to traditional forms of energy, which are susceptible to loss or degradation, exergy retains its inherent value unless there is a transfer of mass or heat. This highlights the significance of energy conservation and waste reduction in economic systems. By underscoring the interconnection between human conduct and the natural environment, thermoeconomics challenges the conventional economic perspective that prioritizes the human element (Tallgauer & Schank, 2024). It recognizes that the availability of natural resources and the capacity of the environment to manage waste impose constraints on economic activity. The integration of physics and ecological concepts in thermoeconomics provides a more comprehensive outlook on the sustainability of economic systems (Arslan & Yilmaz, 2024). This paper proposes a new theoretical framework for systemic sustainability, the exergy-based management model, which integrates thermoeconomics with business science. This novel approach not only underscores the significance of exergy preservation in economic operations but also underscores the necessity for organizational flexibility (Dimple & Tripathi, 2024; Rumman & Alqudah, 2024; Singh et al., 2021a, b). In the context of this theoretical contribution, organizational flexibility is a crucial factor in enabling dynamic adaptation to changing economic conditions, thereby ensuring sustainable and resilient management in the long term (Pandey et al., 2024).

This paper introduces the Systemic Exergy Management (SYMΞX) framework, a novel theoretical model that fuses thermoeconomic principles with business science to create a holistic sustainability assessment tool. In contrast to existing models, SYMΞX is distinctive in its integration of exergy analysis with corporate performance evaluation, thereby enabling organizations to assess financial outcomes and resource efficiency concurrently. This novel approach addresses deficiencies in the existing literature by demonstrating the potential of thermodynamic principles as a means of enhancing corporate performance metrics, thereby facilitating the development of sustainable business practices that are firmly grounded in systemic efficiency (Tsatsaronis, 2011). Similarly, Gong et al. (2018) emphasize the necessity for dynamic sustainability performance frameworks based on real-time data, which can enhance decision-making processes pertaining to environmental, social, and governance (ESG) initiatives. Moreover, for readers less familiar with thermoeconomics, it is essential to provide clarification regarding the key terms utilized throughout this manuscript. The term "exergy" is used to quantify the potential for useful work within an energy flow, whereas "entropy" is employed to indicate the extent of disorder in a system. This represents the energy losses that are unavailable for productive work. The SYMΞX framework is founded upon thermodynamic principles, which enable it to assess resource efficiency and sustainability.

## Research Design

This study adopts a theoretical building approach that uses abductive inference to integrate concepts from thermodynamics and management science (Rivard, 2021). This method entails the construction of a novel theoretical framework through the integration of concepts and principles drawn from disparate disciplinary domains. In this instance, the integration of thermodynamics, which examines the transformation of heat, and management science, which concerns the organization and control of activities, resulted in the development of a novel conceptual framework. This framework was utilized to investigate the potential applications of thermoeconomics in the domain of business performance measurement (Vila-Henninger et al., 2022).

The initial phase of the study entailed a comprehensive examination of pertinent scholarly literature. A variety of scholarly books and papers from disparate fields, including economics, sustainability studies, management science, and thermodynamics, were subjected to examination. The principal objectives of this research were to identify the fundamental concepts and theories pertaining to thermoeconomics, sustainability reporting, and corporate performance assessment. This review formed the basis for the development of several conceptual propositions (Lager & Simms, 2023). The objective of these propositions was to establish a link between the potential contributions of thermoeconomics to the field of sustainability management and the performance assessment frameworks currently in use. Subsequently, an abductive reasoning process was employed, following the completion of the literature review. Abductive inference is an iterative process that involves moving between empirical observations and theoretical explanations in order to refine and construct new theoretical frameworks (Shepherd & Suddaby, 2016). The study integrated the developed conceptual propositions through this process, resulting in a novel theoretical abstraction. This abstraction serves as a framework for incorporating thermoeconomics principles into corporate performance measurement.

### Structure of the Work

The following is a description of the organization of the article. In order to establish the fundamental principles of business science and thermoeconomics, the section entitled "Research Design" presents a comprehensive review and analysis of the essential literature in the field. The theoretical building process and research analytical tools are elucidated in Section "Theoretical Background." The Systemic Exergy Management (SYMΞX) framework is elucidated in Section "Systemic Exergy Management (SYMΞX)," wherein its fundamental tenets and practical applications are also expounded. The impact of SYMΞX on management theories is investigated in the "Contribution to the Development of Management Theories" section, with a particular emphasis on its capacity to be integrated seamlessly with existing paradigms and its suitability for bridging model gaps. In the section entitled "Enhancing Organizational Flexibility Through SYMΞX," real-world examples are provided to illustrate the way the novel framework is applied to a variety of business tasks. The primary findings are summarized in the "Conclusion" section, accompanied by a discussion of potential policy recommendations.

## Theoretical Background

In the contemporary historical period, when the impact of human activity on the planet is more evident than ever, it is imperative to adopt a comprehensive approach to sustainability (Peng et al., 2021). The classic three pillars of sustainability (economy, environment and society) have long provided a framework for understanding and addressing the challenges of sustainable development (Srivastava et al., 2022) including through the sustainable development goals (SDGs) (Colasante et al., 2024). However, the accelerated evolution of global systems has rendered the conventional approach to analysis inadequate. A more integrated perspective that transcends traditional boundaries is necessary to comprehend the rapidly changing complexity of these systems. This necessitates a shift in focus from proximate economic advantages to a more comprehensive examination of the long-term consequences of organizational and individual actions on the natural environment and social systems (Piccarozzi et al., 2022). The application of innovative approaches to sustainability knowledge, such as thermoeconomics, can facilitate a comprehensive assessment of the sustainability of processes and systems. This approach ensures that the ability of future generations to meet their needs is not compromised (Ahmad et al., 2023). Thermoeconomics, which combines the principles of thermodynamics with economic theory, offers a unique perspective on sustainability (Valentinov, 2021). The application of thermoeconomics enables the accurate assessment of the true costs associated with economic activities. This is achieved by the quantification of energy flows and the identification of exergy losses. In contrast to conventional economic analysis, which frequently concentrates exclusively on financial costs, thermoeconomics takes into account environmental, social, and resource depletion expenses. For example, consider an industrial process that generates high levels of air pollution and public health impacts (Wang et al., 2022) but also on buildings (Ruffolo et al., 2023); thermoeconomics can reveal the significant energy costs associated with this pollution and its social impacts, providing a more accurate assessment of the total cost of production (Fu & Su, 2021). The integration of complex systems thinking into sustainability studies is a crucial endeavor, as it enables the observation of the interconnectivity of all elements within a system and the comprehension of how alterations in one component can precipitate ripple effects across the entire system (West et al., 2020). This perspective is essential for the development of strategies that are not only effective in the short term but also sustainable in the long term.

## The Contribution of Thermoeconomics to Management Studies

The field of thermoeconomics has the potential to make a substantial impact on the field of management science. By providing a framework for integrating economic analysis with the fundamental principles of thermodynamics, particularly the concept of exergy, thermoeconomics offers a unique opportunity to advance our understanding of complex systems. A fundamental concept in thermodynamics, exergy is defined as the maximum useful work that can be extracted from a system in each state, with the environment serving as a reference point. It represents the potential for a system to perform work while maintaining equilibrium with its surrounding environment. In essence, exergy is a measure of the quality of energy, distinguishing between high-quality energy, which can be readily converted into useful work, and low-quality energy, which is difficult to utilize effectively (Catrini, Cipollina, et al., 2018). This integration facilitates a more comprehensive understanding of the efficiency and sustainability of production and consumption processes. In the field of management science, thermoeconomics can be employed to optimize the design and operation of energy systems, improve resource allocation, and enhance environmental performance (Liu & Shen, 2022) and increased flexibility for businesses (Litvinova et al., 2023). In the contemporary business environment, characterized by rapid change and uncertainty, flexibility is a crucial attribute for organizations to possess. The ability to adapt to shifts in resource availability, energy costs, and environmental regulations is imperative for businesses to survive and thrive (Yuan & Zhang, 2020). For instance, thermoeconomic analysis can assist in the identification of the most efficient methodologies for the utilization of energy and materials in production processes, which may result in cost savings and a reduction in environmental impact (D'Adamo et al., 2023). It can also provide insights into the true economic value of goods and services, considering the energy and material inputs required for their production (Casisi et al., 2023). In addition, thermoeconomics can contribute to the development of sustainable business models (Barletta et al., 2024) that consider the long-term availability of resources and the environmental consequences of business activities (Ja'fari et al., 2023). By assessing the energy consumption of different processes, managers can make more informed decisions that are in line with sustainability goals (Karthikeyan & Praveen Kumar, 2023). The theoretical relationship between thermoeconomics and management can be summarized in the following key points:

- **Efficiency and Exergy:** The concept of exergy, which represents the maximum useful energy that can be extracted from a system for performing valuable tasks, is a central tenet of thermoeconomics. This concept is closely aligned with the principles of efficiency in management, which seek to maximize output while minimizing input resources. By identifying and reducing exergy losses, managers can optimize resource utilization and enhance overall system efficiency (Ekrataleshian et al., 2021).
- **Cost Allocation and Exergoeconomic Analysis:** It is a specialized field that integrates the principles of thermodynamics and economics to optimize energy systems. The assignment of monetary value to exergy allows for a more precise allocation of costs to the various components and processes involved. This approach designated "exergoeconomic analysis," fosters transparency in pricing and facilitates informed decision-making. Consequently, it has become an indispensable instrument for engineers and economists striving to enhance the efficiency and sustainability of energy systems (Aghbashlo & Rosen, 2018).
- **Sustainability and Exergy:** The principles of thermoeconomics are aligned with those of environmental stewardship and long-term economic viability, as they promote the sustainable utilization of energy and resources. By optimizing exergy utilization, managers can reduce waste and conserve valuable resources, thereby contributing to the development of a more sustainable business model**.** (Erzen et al., 2021).
- **Process Optimization and Thermoeconomic Indicators:** The application of thermoeconomic analysis provides a means of identifying and quantifying inefficiencies inherent to industrial processes. By analyzing exergy destruction and losses, managers can identify areas for improvement and optimize process designs, which ultimately leads to reduced costs and increased productivity (Cao et al., 2022).
- **Resource Management and Exergoeconomic Benchmarks:** The application of thermoeconomic benchmarking enables managerial personnel to undertake comparative analyses of the efficiency of disparate processes or systems, thereby furnishing a foundation for the formulation of improvement initiatives. By establishing exergoeconomic benchmarks, organizations can monitor their progress toward optimized resource utilization and cost-effectiveness (Ogorure et al., 2024).

A schematic representation of the concepts is presented in Table 1, which emphasizes the interrelationship between exergy (a measure of the maximum useful work that can be

**Table 1** Relationships between key concepts in thermoeconomics and management science

| Key concepts | Thermoeconomics | Management |
|---|---|---|
| Exergy | Maximum useful work that can be extracted from a system | Efficiency in resource utilization |
| Entropy | Measure of disorder or unavailable energy | Inefficiency in resource utilization |
| Economic efficiency | Ratio of output to input in terms of resources | Ability to achieve organizational goals with minimum resource expenditure |

extracted from a system), entropy (a measure of disorder or randomness), and economic efficiency. This framework establishes links between key management areas, including resource allocation (the process of allocating resources to specific activities), sustainability (meeting the needs of the present without compromising the ability of future generations to meet their own needs), resilience (the ability to recover from perturbations), decision making (the process of selecting the best course of action) and innovation (the introduction of new ideas, products or services).

A review of the key concepts presented in the table reveals that exergy represents the maximum useful work that can be extracted from a system in thermoeconomics. Consequently, it gauges the quality or accessibility of energy, considering both the energy density and the capacity to perform work. In the context of management, energy can be considered analogous to the efficiency of resource utilization, emphasizing the extent to which resources are transformed into valuable outputs. Entropy serves to quantify the degree of disorder or randomness intrinsic to a given system. It thereby indicates the extent of energy that is unavailable for useful work due to the inherent irreversibility and losses inherent to such processes. In the field of management science, it signifies the inefficiency of resources, which ultimately results in waste and a decline in economic efficiency. In the context of management, economic efficiency is defined as the ratio of output to input concerning resources such as energy, materials, or labor. It reflects the effectiveness with which resources are utilized in order to achieve organizational flexibility. The field of thermoeconomics is concerned with aligning economic efficiency with the minimization of exergy degradation and entropy production.

Table 2 applies thermodynamic principles (Ağbulut, 2022) to management, advocating a cross-disciplinary approach to energy exchange. It correlates the fundamental tenets of thermodynamics with a range of managerial, social, technological, and physical interpretations. This comprehensive framework permits the examination of energy efficiency and resource utilization across a range of scenarios. A comprehensive approach is thereby attained, whereby thermodynamic principles are translated into practical strategies for the management of resources, technological innovation, and societal accountability. As illustrated in Table 2, a comprehensive and detailed multidimensional description of the principles of thermodynamics is provided below.

### Principle 0 (Zeroth Law)

- **Physical Interpretation:** The zeroth law of thermodynamics establishes that if two systems are each in thermal equilibrium with a third system, they are also in equilibrium with each other (Honig, 2014).
- **Managerial Interpretation:** Organizations should aim to balance their exergy levels to direct valuable resources toward the most efficient processes, aligning with energy efficiency principles (Naletov et al., 2020).
- **Social Interpretation:** Societies must pursue equitable conditions, ensuring resource access for all individuals, reflecting fairness and justice in resource distribution (Ali & Kamraju, 2023).
- **Technological Interpretation:** Prioritization of technologies that optimize energy extraction and minimize entropy is essential, resonating with resource optimization goals (Williams, 2024).

### Principle 1 (First Law)

- **Physical Interpretation:** The first law of thermodynamics asserts the constancy of total energy in an isolated system over time (Aghion & Green, 2023).
- **Managerial Interpretation:** Organizations should engage in resource conservation and exergy efficiency, ensuring sustainable operations without resource wastage (Dehghani & Yoo, 2020).
- **Social Interpretation:** Societies must embrace environmental stewardship, conserving resources for future generations and minimizing ecological impact (Hassan et al., 2021).
- **Technological Interpretation:** The adoption of technologies that enhance energy efficiency and reduce waste is crucial, reflecting the principles of sustainable technology development (Zevenhoven, 2021).

**Table 2** Managerial interpretation of thermodynamic principles

| Thermodynamic principle | Key concept | Managerial interpretation | Social interpretation | Technological interpretation | Physical interpretation |
|---|---|---|---|---|---|
| Principle 0 (Zeroth Law) | Equivalence of Exergy | Equalize exergy levels | Promote equity and fairness | Utilize efficient technologies | Energy equivalence |
| Principle 1 (First Law) | Exergy Conservation | Conserve resources | Embrace sustainability practices | Optimize resource allocation | Energy conservation |
| Principle 2 (Second Law) | Exergy Degradation | Minimize waste | Reduce negative externalities | Develop innovative processes | Entropy increase |
| Principle 3 (Third Law) | Asymptotic Approach to Entropy | Optimize resource utilization across all conditions | Promote continuous improvement | Adopt adaptive technologies | Entropy maximum at absolute zero |

**Principle 2 (Second Law)**

- **Physical Interpretation:** The second law of thermodynamics dictates that the entropy of an isolated system will invariably increase over time (Neri, 2020).
- **Managerial Interpretation:** Organizations are encouraged to pinpoint and rectify inefficiencies, aiming to reduce exergy degradation and bolster long-term sustainability, in accordance with continuous improvement principles (Fleck, 2023).
- **Social Interpretation:** Societal efforts should focus on curtailing negative externalities like pollution and resource overuse, reflecting a commitment to environmental responsibility (Westbroek et al., 2020).
- **Technological Interpretation:** The development and implementation of innovative processes and technologies that curtail entropy and waste align with waste reduction objectives (Prajapati et al., 2024).

**Principle 3 (Third Law)**

- **Physical Interpretation:** The third law of thermodynamics states that the entropy of a system approaches a maximum value as the temperature approaches absolute zero (Buffoni et al., 2022).
- **Managerial Interpretation:** Organizations should strive for exergy efficiency across the entire range of operating conditions, maximizing resource utilization and minimizing waste generation. This aligns with the concept of wide-range optimization, which emphasizes the need to consider resource efficiency at all stages of the production process (Mimkes, 2017).
- **Social Interpretation:** Societies should strive for continuous improvement in social welfare and equity across all demographics and socioeconomic groups. This aligns with the concept of inclusive development, which focuses on ensuring that the benefits of progress are shared equitably (Tsekov, 2023).
- **Technological Interpretation:** Technologies that are adaptable to various operating conditions and can function efficiently over a wide range of temperatures should be prioritized. This aligns with the concept of universal applicability, which emphasizes the need for technologies that can be used in diverse settings and under varying conditions (X. Chen et al., 2024).

Figure 1 presents a conceptual diagram that elucidates the interrelationships between pivotal concepts in thermoeconomics and management. The diagram's core concepts are exergy, entropy, and economic efficiency, which are represented as interconnected nodes. Exergy represents the maximum useful work that can be extracted from a system, while entropy measures the degree of disorder or unavailable energy. Economic efficiency is defined as the ratio of output to input in terms of resources. The arrows indicate the dynamic interactions between these concepts, thereby demonstrating how thermoeconomics integrates with management principles to optimize resource use, minimize waste, and improve overall system efficiency.

**Theory of Thermoeconomic Management**

The concept of thermoeconomics represents a novel approach that integrates thermodynamic principles with financial considerations, offering a novel system for coordinating business activities toward feasible development (Cardullo & Liu, 2017). By establishing a connection between the physical and financial realms, this unconventional approach aims to optimize financial flows with a focus on asset productivity and natural preservation. The fundamental concept of thermoeconomics is energy, which

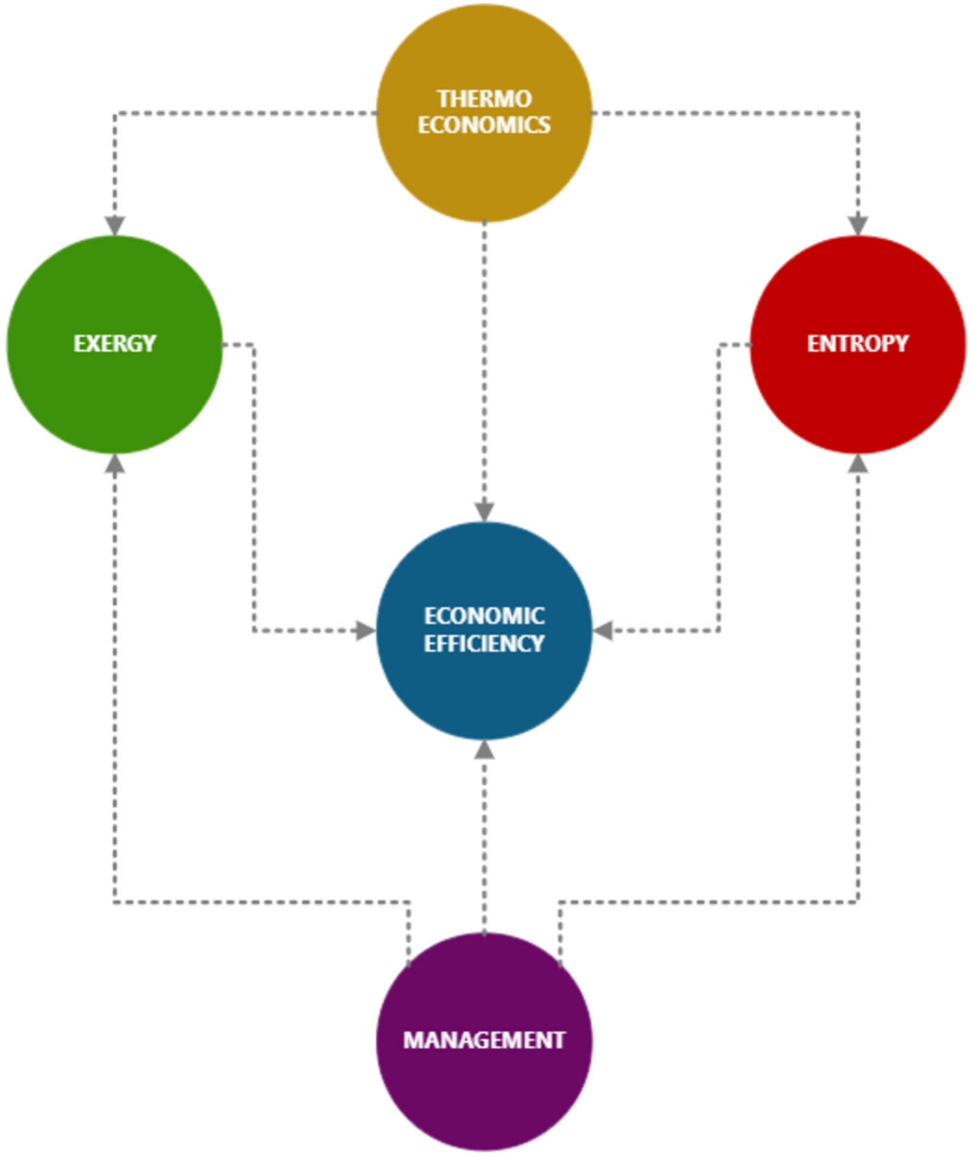


**Fig. 1** Conceptual diagram of Thermoeconomics and Management

can be defined as a thermodynamic quantity representing the potential work that can be extracted from a system. While energy is subject to variation, exergy remains constant in the absence of net mass or heat exchange (Ghaebi et al., 2011). This underscores the critical necessity to curtail energy wastage and conserve energy in business operations. Consequently, thermoeconomics utilizes energy flow analysis to identify inefficient processes and potential avenues for asset preservation (Mahmoudan et al., 2021).

The comprehensive approach adopted by thermoeconomics enables the integration of ecological concepts with the fundamental laws of physics. This perspective asserts that economic activity is inherently constrained by the availability of natural resources (Frangopoulos & Dimopoulos, 2024). Furthermore, thermoeconomics offers a more comprehensive grasp of the complex interconnections between the environmental, social, technological, and economic dimensions of sustainability. It acknowledges the interdependence between human activity and the natural world, challenging the anthropocentric perspective that has traditionally informed economic theory (Sciubba, 2019). It is possible to evaluate the sustainability of economic systems using novel analytical approaches that thermoeconomics may offer. This novel method has the potential to provide decision-makers with new tools for redefining policies that enhance organizational efficiency, reduce environmental effects, and support long-term sustainability. This is achieved by measuring energy flows and accounting for ecological limits (Simpson & Edwards, 2011).

**Proposition 1** (P1): *Thermoeconomics, a convergence of thermodynamics and economics, offers a holistic approach to sustainable development that optimizes resource use and minimizes environmental impact.*

The field of thermoeconomics extends beyond the conventional boundaries of economic study. The fundamental ideas of thermoeconomics can be applied at various economic levels, including the mesoeconomic (industry) and microeconomic (business) levels, despite its initial development at the macroeconomic level (Chamchine et al., 2006). It is conceivable that thermoeconomics may prove to be an efficacious instrument at the meso-economic level, capable of influencing the entirety of a company toward the adoption of sustainable practices. Those in positions of authority within the industry, in conjunction with policy officials, are able to identify sectors where there is a notable decline in energy efficiency by undertaking an examination of the energy flows within specific industries (Dincer, 2002). The data presented herein can facilitate the implementation of targeted initiatives, including the promotion of energy-efficient technology and the facilitation of industrial collaboration aimed at the development of optimal resource utilization procedures (Rosa & Beloborodko, 2015). From an industrial perspective, this strategy has the potential to markedly diminish overall energy utilization and environmental impact. Similarly, thermoeconomics enables businesses to make data-driven decisions that improve their sustainability on a microlevel. The application of exergetic analysis enables businesses to identify inefficiencies in their manufacturing processes. Such an approach can identify potential avenues for enhancement, including the selection of raw materials with a higher embodied energy (thereby necessitating less processing) or the optimization of heating or cooling systems (Caglayan & Caliskan, 2022). By reducing energy degradation in their operations, businesses may achieve significant cost savings and environmental benefits. It is possible to achieve a more feasible item life cycle if the organization adopts the use of lighter, more energy-intensive materials that utilize less energy during the manufacturing process. The thermoeconomics may facilitate a cascading impact of beneficial financial and natural outcomes by comprehending and optimizing the flow of assets and energies at each level, from small businesses to entire divisions. This multi-level approach guarantees that asset effectiveness and natural obligation become an integral component of all financial operations, thereby providing a comprehensive framework for achieving economic advancement (Calise et al., 2006).

**Proposition 2** (P2): *By integrating the laws of physics with economic principles, thermoeconomics can understand and optimize the flow of resources and energy and help guide decision-making toward more sustainable outcomes.*

A novel approach to product creation and marketing strategies is given by thermoeconomics (Cabello, 2022). Designers can constrain energy degradation and entropy ordering in their products, minimizing their impact on the environment and optimizing resource utilization (Abadías Llamas et al., 2020). This is frequently achieved by examining the thermodynamic aspects of materials and energy flows. Furthermore, thermoeconomic analysis can facilitate the selection and development of materials and production methods that minimize energy consumption and environmental impact, ultimately resulting in the creation of more environmentally conscious products (Wu et al., 2021). Furthermore, thermoeconomics can inform marketing plans by emphasizing a product's efficiency and sustainability. The application of thermal economic concepts allows companies to demonstrate the environmental benefits of their products, including reduced energy consumption, more efficient use of resources, and the minimization of waste (Rosa & Beloborodko, 2015). It is possible for businesses to gain a competitive advantage by targeting consumers who are environmentally concerned with marketing messages that are aligned with sustainability ideals. Moreover, thermoeconomics facilitates the traceability and transparency of supply chains, which enables businesses to effectively communicate the product's sustainable credentials (Catrini et al., 2018a, 2018b).

**Proposition 3** (P3): *Thermoeconomics innovates product design, manufacturing, and supply chain management to promote resource efficiency, sustainability, and consumer engagement.*

Financial measures such as revenues, profit margins, and return on investment have historically been employed to evaluate the performance of business entities (Tasáryová et al., 2021). Notwithstanding, these indicators are inadequate for encompassing the full spectrum of organizational benefits, particularly with respect to resource efficiency, social responsibility, and environmental sustainability (Imeni & Edalatpanah, 2023). The application of thermoeconomics facilitates a comprehensive understanding of organizational performance by integrating thermodynamic principles into performance measurement systems, thereby addressing this constraint (Li, 2016). Exergy-based accounting represents a novel statistical approach that can be integrated into thermoeconomics. It assesses the efficiency with which resources are utilized within a company, evaluating both the effectiveness of resources and their impact on the environment, in addition to the financial value generated by organizational operations (Rocco et al., 2014). The integration of exergetic analysis into performance evaluation allows businesses to identify inefficiencies in the usage of resources and develop sustainable performance (Haydargil & Abuşoğlu, 2018).

**Proposition 4** (P4): *Thermoeconomics promises to transform business management by introducing metrics that capture economic, environmental, and resource efficiency, incentivize sustainability, and improve organizational performance.*

Using propositional logic (Steen & Fuenmayor, 2024), the following theoretical statement can be made based on the previous propositions:

$T1\,(Overall\,Theory\,1)\;=P1\wedge P2\;\rightarrow\;(P3\wedge P4)$

where:

1. **T1 (Overall Theory 1):** represents the overall theory we are building.
2. **P1 ∧ P2:** This conjunction underlies this part of the theory. It states that thermoeconomics (P1) offers a holistic approach to sustainability (P1) by integrating thermodynamics (understanding resource flows) with economics (decision-making) (P2). This combined approach allows for optimizing resource use and minimizing environmental impact (P1).
3. **P3 ∧ P4:** This conjunction represents the potential outcomes of applying thermoeconomics in business management. It suggests that thermoeconomics can lead to:

    - **P3:** Innovation in product design, manufacturing, and supply chain management, fostering resource efficiency, sustainability, and consumer engagement. By understanding resource flows and their environmental impact, thermoeconomics guides development toward sustainable practices and engages consumers who value sustainability.

    - **P4:** Transformation of business management by introducing metrics that capture not only economic performance but also environmental and resource efficiency. These metrics incentivize sustainability within the organization and contribute to improved overall performance.

In light of these considerations, T1 underscores the pivotal role of a holistic approach (P1) and an understanding of resource flow (P2) in achieving both innovation in sustainability practices (P3) and a transformation of business management (P4).

## Thermoeconomic Management Across Dimensions

As we navigate the intricate terrain of contemporary management, it is evident that conventional strategies for maximizing profit within market limitations are no longer adequate. The urgent issues of environmental degradation and resource depletion demand a fundamental transformation of economic paradigms (Spash, 2020). The concept of exergy, a pivotal yet frequently overlooked aspect of energy, serves as the foundation for this transformation. The objective of traditional management has consistently been the maximization of efficiency, or the achievement of greater results with the same or fewer resources (Martínez-Alonso et al., 2020). But this approach often ignores the highly valuable component of energy available for work: exergy (Grainger & Smith, 2021). This oversight has significant implications for the field of thermodynamics, as any transformation process will eventually result in a decrease in exergy. This phenomenon signifies the loss and waste of valuable work potential in the form of heat (Ustaoglu et al., 2022). The importance of understanding and addressing exergy losses in organizational processes is underscored by this omission. By implementing this approach, we can reduce our environmental impact and optimize resource utilization. By prioritizing sustainability and conscientious resource management, it is possible to develop a management style that is more effective, efficient, and environmentally friendly, with a positive impact on both the environment and the business. This shift in perspective can be achieved by adopting the concept of exergy as a foundational principle (He et al., 2020). It is of the utmost importance to recognize that exergy serves as a powerful operational instrument, capable of addressing real-world issues both in theory and practice. By adopting a more holistic perspective, we can ensure that future generations enjoy a more sustainable future.

**Proposition 5** (P5): *Traditional profit-maximizing approaches to management are inadequate in the face of resource depletion and environmental concerns. Embracing exergy can optimize resource use and minimize environmental impact, promoting sustainability and responsible resource management.*

The concept of exergy represents a fundamental tenet in thermoeconomics, offering a means to ascertain the intrinsic value of our energy resources. Exergy represents the measure of the usable work potential inherent within an energy flow, as opposed to energy, which is inherently diminished with each change (Rúa Orozco et al., 2024). The concept is significant in that it illustrates the true value of a resource when it is utilized for the achievement of meaningful objectives. For businesses seeking to be responsible custodians of their resources, reducing energy consumption represents a significant challenge (Bernat et al., 2022). The conventional approach to enhancing efficiency typically concentrates on curbing energy usage. However, in the context of today's constrained resource scenario, this strategy is inadequate (Wohlfarth et al., 2020). By employing energy analysis, organizations can identify areas where critical resources are being expended ineffectively or in a wasteful manner. This analytical approach offers valuable insight into the utilization of resources within a given system (James et al., 2020). The implementation of exergy analysis enables organizations to optimize the utilization of their resources, reduce waste, and extend the lifespan of essential resources (Nikam et al., 2021). By reducing energy losses across their operations, this strategy also assists businesses in mitigating their environmental impact (G. Singh et al., 2021a, 2021b). Energy loss is a pervasive phenomenon that occurs at each stage of the production process, frequently resulting in the generation of waste heat and the depletion of resources (Gupta et al., 2022). By identifying instances of resource misuse or waste, exergy analysis enables companies to achieve financial savings and optimize resource utilization, thereby gaining a competitive advantage in their market (Naderi et al., 2021). For companies striving to optimize their resource utilization, mitigate their environmental impact, and maintain competitiveness in a context of constrained resources, it serves as a highly effective instrument (Brändström & Eriksson, 2022).

**Proposition 6** (P6): *Exergy, a pivotal concept in thermoeconomics, uncovers energy resource value. Addressing exergy loss boosts resource efficiency, cuts waste, and lowers environmental impact, enhancing organizational flexibility and business competitiveness.*

The implementation of an innovative management strategy founded upon thermoeconomic principles is imperative for the effective resolution of these issues. It is imperative that this strategy transcends the conventional approaches to sustainability that have been employed thus far and concentrates on maximizing resource efficiency throughout the entire value chain (Fernandes et al., 2020). This strategy should transform organizational procedures and decision-making, guided by the idea of exergy (Fernandes et al., 2020). A set of fundamental concepts that facilitate our understanding of energy and its role in optimizing resources form the basis of thermoeconomics. The principle of exergy equivalence serves to illustrate that the total exergy within a closed system remains constant (Romero & Linares, 2014). This suggests that, while energy cannot be generated or destroyed, it can undergo a transformation in form. For businesses seeking to monitor energy flows throughout their operations and allocate resources in an optimal manner, this principle is of paramount importance. The second principle, exergy conservation, emphasizes the minimization of exergy degradation. It is inevitable that a certain amount of energy will be converted into useless heat during a process. The objective of thermoeconomics is to reduce this energy loss by streamlining procedures and advancing technology (Chen et al., 2024). It is possible for organizations to reduce waste and make optimal use of their valuable resources by following this approach. The third principle, the Asymptotic Approach to Entropy, acknowledges that all systems ultimately reach a maximum entropy state from which no further usable work can be derived. In order to reduce energy deterioration and reach a condition closer to optimal utilization, thermoeconomics, on the other hand, promotes a never-ending quest of improvement, pushing firms to aim for constant innovation and process optimization (Dobija & Renkas, 2021). In conclusion, the concept of systemic sustainability underscores the necessity for a comprehensive approach to sustainability that integrates the technological, social, economic, and natural aspects of sustainability (Khlif et al., 2022). It is possible for organizations to make decisions that are beneficial to society, the environment, and their financial line of business when they take these dimensions of energy flows into account. This strategy recognizes that sustainability necessitates the attainment of a delicate equilibrium between these competing objectives, encompassing both environmental conservation and economic growth (Gabbi et al., 2021).

**Proposition 7** (P7): *The total exergy within a system remains constant, and exergy can be transformed but not created or destroyed, guiding organizations to optimize resource allocation and minimize waste.*

**Proposition 8** (P8): *Continuous innovation and process optimization are necessary to minimize exergy degradation and achieve a state closer to perfect utilization.*

**Proposition 9** (P9): Minimizing exergy degradation is crucial for maximizing resource utilization and achieving systemic sustainability, balancing environmental, social, technological and economic interests.

In light of the aforementioned analysis, it can be posited that thermoeconomic management (TM) represents a strategy (Kopnina, 2017) that seeks to integrate the fundamental principles of thermodynamics (Varbanov et al., 2020) into the very core of business operations (Nadi & Górnicki, 2022). This integration is undertaken with the objective of effecting a transformation in our conceptualization of sustainability (Abeysekera, 2022). This innovative method provides a comprehensive framework for maximizing resource use and reducing environmental impact in all aspects of a business. Adoption of the thermoeconomic approach allows businesses to optimize resource utilization, modify operational procedures, and reduce their ecological impact in a substantial and quantifiable manner (Jiménez & Baena, 2024). It can be argued that thermoeconomic management (TM) represents an effective technique for maximizing energy efficiency and reducing waste in the physical realm (thermophysics) (Clairand et al., 2020). TM identifies areas of inefficiency and directs interventions to enhance energy efficiency and mitigate energy losses by examining energy flows within organizational activities. By determining the exergy content of resources and directing activities to maximize their use while minimizing their depletion, TEM also increases resource efficiency (Llamas et al., 2024). T This entails the implementation of tactics such as material recycling, reuse, and waste-to-resource conversion, which utilize the energy content of waste streams to extract additional value. By employing exergy analysis, TM facilitates decision-making processes within the management dimension (thermobusiness) by providing managers with insight into the exergetic performance of various solutions, thereby enabling well-informed choices that prioritize waste minimization and resource efficiency (Selicati & Cardinale, 2021). Moreover, the integration of energy considerations into organizational planning facilitates the development of sustainable business models and strategies for businesses. This allows them to align their operations with the values of environmental stewardship and resource conservation. Within the social dimension (thermosocial), TM fosters an organizational culture of resource conservation and environmental responsibility, thereby promoting employee well-being and engagement in sustainability projects (Arias et al., 2023). TM facilitates the integration of broader social concerns into the decision-making processes of stakeholders by enhancing their awareness of the social implications of business actions and products (Cano-Londoño et al., 2022). In addition, TM guides the implementation of energy-efficient technologies and processes within the thermo-technological domain. This is achieved by evaluating their exergetic performance and identifying areas for improvement (Silva et al., 2020).

**Proposition 10** (P10): *Thermoeconomic management optimizes energy use and minimizes waste by analyzing exergy flows, guiding interventions for improved efficiency and resource utilization.*

**Proposition 11** (P11): *Thermoeconomic management informs decision-making through exergy analysis, enabling resource-efficient and waste-minimizing choices, fostering sustainable business models.*

**Proposition 12** (P12): *Thermoeconomic management promotes employee engagement in sustainability by fostering resource conservation and awareness of social impacts.*

**Proposition 13** (P13): *Thermoeconomic management guides adoption of energy-efficient technologies, development of waste-to-resource solutions, and integration of renewable energy sources.*

Based on the previous proposition, the following theoretical statement can be made:

$$T2\,(Overall\,Theory\,2) = P5 \wedge (P6 \vee P7) \rightarrow (P8 \wedge P9) \rightarrow (P10 \wedge P11 \wedge P12 \wedge P13)$$

The proposition put forth by this theory is that thermoeconomic management (TM) provides a novel framework for attaining sustainable business practices. This is achieved by establishing a connection between the conventional objective of profit maximization and the realities of resource depletion and environmental concerns. The location of the aforementioned point is as follows:

1. **P5 ∧ (P6 ∨ P7):** This conjunction states that the limitations of traditional approaches (P5) become more critical when considering either the value of resources highlighted by exergy (P6) or the principle of exergy conservation (P7).
2. **P8 ∧ P9:** This conjunction emphasizes the ongoing pursuit of minimizing exergy degradation (P8) as essential for achieving systemic sustainability, balancing environmental, social, technological, and economic aspects (P9).
3. **P10 ∧ P11 ∧ P12 ∧ P13):** This conjunction demonstrates the practical applications of TEM across various dimensions:
   - **P10:** Optimizing resource use and minimizing waste through exergy analysis.
   - **P11:** Informing decision-making for better resource efficiency and waste minimization, ultimately promoting sustainable business models.
   - **P12:** Fostering employee engagement in sustainability initiatives.
   - **P13:** Guiding the adoption of sustainable technologies and practices.
4. **The first implication " → "** connects the limitations of traditional approaches (P5) with the need for minimizing exergy degradation (P8 & P9). It suggests that if traditional approaches are inadequate (P5 is

true), then organizations must consider minimizing exergy degradation (P8 & P9) to achieve sustainability.

5. **The second implication " → "** connects minimizing exergy degradation (P8 & P9) with the practical applications of TM (P10-P13). It suggests that if organizations strive to minimize exergy degradation (P8 & P9), then they can leverage TM practices (P10-P13) for resource efficiency and sustainability.

T2 theory suggests that when the value of resources is assessed in terms of exergy (P6) or the concept of energy conservation (P7), the limitations of conventional profit maximization approaches (P5) become more apparent. Organizations are compelled to pursue relentless innovation and process optimization (P8) in order to achieve optimal resource utilization (P8 & P9). This is achieved by acknowledging the constraints of resources and the potential for waste minimization. By employing TEM (P10-P13), the objective of minimizing energy degradation can be effectively translated into practical applications across a range of business-related domains.

## Thermoeconomic Management as a Framework for Systemic Sustainability

An integrated strategy that considers sustainability's social, technological, economic, and environmental aspects is defined as systemic sustainability (Tong et al., 2018). By recognizing the interconnectivity of these dimensions and the necessity of holistic solutions in complex situations, this approach transcends the limitations of conventional sustainability strategies (Villeneuve et al., 2017). The reduction of deleterious effects on the environment, the safeguarding of natural resources, and the mitigation of climate change are integral aspects of environmental sustainability (Pascarella & Bednar, 2021). The primary objective of economic sustainability is to achieve long-term viability and prosperity while maintaining a balance between economic growth, social progress, and environmental preservation (Borgida et al., 2022). The development and implementation of environmentally friendly technology and practices are exemplified by technological sustainability (Vacchi et al., 2024). The objective of social sustainability is to advance social justice, inclusivity, and community well-being while ensuring the equitable distribution of economic and environmental advantages throughout society (Nilsson et al., 2024).

**Proposition 14** (P14): *Systemic sustainability integrates environment, economy, technology and society, recognizing their interconnectedness for holistic solutions to complex challenges.*

A framework for achieving system-wide sustainability can be established through the implementation of exergy analysis and thermoeconomic management (TM) (Walker et al., 2021). An exergetic analysis is a method of assessing the potential for useful work in energy flows. It adopts a holistic perspective that goes beyond the scope of traditional energy-saving measures (Selicati et al., 2022). By quantifying the potential for productive work in resource flows, TM draws attention to the economic and environmental consequences associated with manufacturing processes (Torres et al., 2020). The integration of environmental and economic considerations facilitates the identification of knowledge gaps and enhances comprehension (experimental analysis). By identifying the process areas where energy degradation occurs, TM facilitates the adoption of resource-saving processes and technologies (Heydari, 2022). This minimizes negative environmental impacts while providing financial benefits and cost savings (process optimization) (De Falco & Capocelli, 2020). TM encourages a comprehensive examination of energy flows throughout the product life cycle from a broad perspective. This strategy is designed to advance sustainability by integrating social, environmental, and economic considerations, including responsible sourcing practices, throughout the product life cycle, from the initial extraction of resources to the end of the product's life cycle (life cycle perspective) (Selicati et al., 2021).

**Proposition 15** (P15): *Thermoeconomic management leverages exergy analysis for systemic sustainability, quantifying resource potential, optimizing processes, and integrating social & environmental factors across a product's life cycle.*

It is not uncommon for conventional management techniques to be implemented in a siloed manner, with an emphasis on the optimization of discrete departments, such as finance and production (Ghita et al., 2021). The promotion of cross-functional collaboration is a key mechanism through which TM can facilitate the dismantling of silos. It is imperative that departments including engineering, procurement, and logistics collaborate to analyze exergy flows across the value chain (Hernandez & Cullen, 2016). Moreover, system-wide optimization represents the primary focus of TM. In contrast to an emphasis on process optimization, TM encourages an understanding of the way disparate processes impact the utilization of resources and the deterioration of energy. In addition, TM incorporates sustainability into its core operations. The integration of exergy analysis and TM principles into decision-making at all levels serves to elevate sustainability from a mere standalone project to an essential component of corporate strategy (Fenerich et al., 2017).

**Proposition 16** (P16): *Thermoeconomic management breaks down silos, promoting cross-functional collaboration for system-wide optimization and integrating sustainability into core business functions.*

The decision-making process surrounding the adoption of new energy-efficient technologies, including the calculation of initial expenses, can often be challenging when it comes to sustainability projects (Hrovatin et al., 2016). TM plays a pivotal role in this intricate balancing act when evaluating energy impact. It would be prudent for organizations to refrain from a total exergy drop throughout the system by conducting a thorough examination of the impact of various decisions on exergy (Huang et al., 2022). Moreover, TM illuminates synergies, the potential for alterations in one domain to engender a beneficial ripple effect in other domains (Chaturvedi et al., 2023). To illustrate, the implementation of water-saving technology may lead to a reduction in energy consumption for water treatment and water conservation, which would have a beneficial impact on both the economy and the environment.

**Proposition 17** (P17): *Thermoeconomic management facilitates trade-off analysis through exergy and reveals synergies across sustainability dimensions.*

Based on the previous proposition, the following theoretical statement can be made:

$$T3\,(Overall\,Theory\,3) = P14 \rightarrow (P15 \wedge P16 \wedge P17)$$

As T3 notes, the integration of traditional, siloed management systems with the interwoven complexity of environmental, economic, technical, and social elements represents a significant paradigm shift in the pursuit of systemic sustainability. This approach, known as thermoeconomic management (TM), bridges the gap between disparate management systems and the intricate interconnections between various factors, offering a powerful framework for achieving sustainability. The location of the aforementioned is as follows:

1. **P14**: This proposition establishes the foundation—systemic sustainability requires considering the interconnectedness of environmental, economic, technological, and social aspects (environment, economy, technology, and society).
2. **P15, P16, and P17**: These propositions represent the key contributions of TM to achieving systemic sustainability:
   - **P15 (TM Leverages Exergy Analysis):** By quantifying resource potential, optimizing processes, and integrating social and environmental factors, TM provides a comprehensive approach that goes beyond traditional methods.
   - **P16 (TM Breaks Down Silos):** Traditional management often operates in silos, hindering a holistic view. TM fosters collaboration and integrates sustainability into core business functions.
   - **P17 (TM Facilitates Trade-offs and Reveals Synergies):** Sustainability decisions often involve trade-offs. TM facilitates informed choices by analyzing exergy impacts and revealing synergies across sustainability dimensions.
3. **The implication "** $\rightarrow$ **"** in T3 signifies a logical connection based on truth values. It suggests that when systemic sustainability requires interconnectedness, TM offers a potential path toward achieving it, but doesn't guarantee success on its own.

## Systemic Exergy Management (Symξx)

Propositional logic was employed as a representation of the theoretical underpinnings of thermoeconomics in connection to exergy, as indicated by the conceptual analysis conducted in the previous section (Steen & Fuenmayor, 2024). In this stage, the conceptual constructs are integrated to establish the theoretical basis for the Systemic Exergy Management (SYMΞX) approach, which serves as a framework for conducting systemic sustainability assessments.

### Theoretical Contribution

SYMΞX goes beyond traditional approaches because it focuses on maximizing the usable work potential (exergy) in the processes taking place in an operating system. The theory underlying SYMΞX is presented below, using propositional logic:

$$\begin{aligned} T\_SYM\Xi X\,(Overall\,Theory) = {} & (P1 \wedge P2) \wedge (P5 \wedge P6) \\ & \rightarrow (P8 \wedge P9 \wedge P14) \\ & \rightarrow (P15 \wedge P16 \wedge P17 \wedge P3 \\ & \quad \wedge P4) \end{aligned}$$

*Explanation:*

1. **Synergy from Thermoeconomics (P1 and P2):** SYMΞX builds upon the holistic approach of thermoeconomics (P1) with a focus on understanding resource flows through exergy analysis (P2). This synergy is crucial for achieving the following outcomes.

2. **Optimizing for Sustainability (P5 and P6):** SYMΞX acknowledges the limitations of traditional profit maximization (P5) and emphasizes exergy as a tool to optimize resource use (P6), promoting sustainability and responsible resource management.
3. **Minimizing Exergy Degradation (P8 and P9):** SYMΞX prioritizes continuous innovation and process optimization (P8) to minimize exergy degradation, leading to maximized resource utilization and systemic sustainability (P9) that considers environmental, social, technological, and economic aspects.
4. **Systemic Transformation through SYMΞX (P15, P16, P17):** By leveraging exergy analysis, SYMΞX facilitates:

   a. Quantifying resource potential, optimizing processes, and integrating social & environmental factors across a product's life cycle (P15).
   b. Breaking down silos and promoting cross-functional collaboration, integrating sustainability into core business functions (P16).
   c. Analyzing trade-offs and revealing synergies across sustainability dimensions (P17).

5. **Sustainable Business Transformation (P3 and P4):** Ultimately, SYMΞX fosters innovation in product design, manufacturing, and supply chain management (P3), leading to the transformation of business management through metrics that capture resource efficiency and incentivize sustainability practices (P4).
6. **The first implication " → "** If both P1 and P2 are true (synergy from thermoeconomics), and both P5 and P6 are true (optimizing for sustainability), then P8, P9, and P14 must also be true (minimizing exergy degradation and achieving systemic sustainability).
7. **The second implication " → "** If P8, P9, and P14 are true (minimizing exergy degradation and achieving systemic sustainability), then P15, P16, P17, P3, and P4 must also be true (systemic transformation and sustainable business practices).

In the following text, the authors present their opinions and knowledge regarding the potential of SYMΞX in the field of management science. In the corporate domain, SYMΞX can act as a catalyst for systemic sustainability by offering a robust theoretical framework that helps overcome the limitations of traditional models. In essence, SYMΞX represents a paradigm shift from the conventional profit-centric approach (Dyck et al., 2023) of thinking with a value-for-money approach (Karikari Appiah et al., 2022). He acknowledges the potential risks associated with an emphasis on immediate financial gain, which can lead to detrimental long-term consequences. Instead, he advocates for an incentive-based approach that encourages sustainable practices. The aforementioned change encourages organizations to optimize resource efficiency by fine-tuning their processes. To mitigate energy losses, SYMΞX also advocates for continuous innovation and development. This underscores the importance of continuous evaluation and development, motivating companies to refine and strengthen their procedures to prevent resource depletion and reduce their environmental impact. Furthermore, SYMΞX mitigates operational impediments and cultivates collaboration through the integration of sustainability into foundational corporate strategies and the promotion of interdepartmental cooperation. Furthermore, SYMΞX is adept at facilitating well-informed decision-making through comprehensive trade-off analysis, which is an additional crucial characteristic. SYMΞX provides companies with the data and resources necessary to make informed decisions regarding emission reduction, as it can quantify the impact of emissions and comprehending the interactions between various sustainability factors. SYMΞX advocates for an all-encompassing approach that considers the intricate interplay between social, technological, economic, and environmental considerations. SYMΞX places a premium on the optimization of resources, which serves to advance technical growth, environmental health, and economic stability.

A concise overview of the Systemic Exergy Management (SYMΞX) conceptual framework can be found in Table 3. The theoretical framework is based on the complementary nature of thermoeconomics and exergy analysis, which together provide the fundamental ideas and constituent parts of this methodology. The SYMΞX methodology is designed to achieve a number of key objectives, including the optimization of sustainability, the minimization of energy degradation, and the realization of systemic transformation. The fundamental tenets underscore the necessity for resource optimization, continuous innovation and enhancement, interdepartmental collaboration, sustainable corporate transformation, and comprehensive trade-off analysis to facilitate well-informed decision-making. The anticipated results of this methodology are a catalyst for systemic sustainability, a shift in mentality away from profit-centeredness to prioritize resource efficiency, ongoing innovation to lessen the impact on the environment and degradation of resources, cooperative efforts to incorporate sustainability into business strategies, and well-informed decision-making through trade-off analysis. To advance sustainability and foster environmental health, economic stability, and technology advancement for a sustainable future, SYMΞX encourages a holistic approach that acknowledges the interplay between environmental, economic, technical, and social variables.

**Table 3** Interconnected elements of systemic exergy management (SYMΞX)

| Theoretical foundations |
| --- |
| Thermoeconomics (P1) + Exergy analysis (P2) |

| Core principles | Key components | Goals |
| --- | --- | --- |
| Synergy from Thermoeconomics (P1 and P2) | Resource optimization | Maximize resource efficiency |
| Optimizing for Sustainability (P5 and P6) | Perpetual innovation and enhancement | Minimize exergy |
| Minimizing Exergy Degradation (P8 and P9) | | Degradation |
| Systemic Transformation through SYMΞX (P15, P16, P17) | Ongoing assessment and advancement | Promote systemic |
| | | Sustainability |
| Sustainable Business Transformation (P3 and P4) | Interdepartmental Collaboration | Foster technological Advancement |
| | Comprehensive Trade-off analysis | Advance Environmental health and economic Stability |
| | Informed decision-making | Complexity |
| | | Organizational flexibility |

| *Outcomes* |
| --- |
| Catalyst for systemic sustainability: SYMΞX surpasses conventional models, prioritizing resource optimization over immediate financial gains |
| Shift in mindset: Moves away from profit-centric thinking to prioritize resource efficiency |
| Continuous innovation: Advocates for ongoing improvement to reduce resource depletion and environmental impact |
| Collaborative efforts: Dismantles operational silos, integrates sustainability into business strategies, and fosters interdepartmental cooperation |
| Informed decision-making: Equips organizations with tools for comprehensive trade-off analysis to diminish overall exergy degradation |
| Holistic approach: Acknowledges interplay between environmental, economic, technological, and social factors |
| Advancing sustainability: Promotes environmental health, economic stability, and technological advancement for a sustainable future |

## Theoretical Framework

Considering the aforementioned concepts and principles derived from abductive inference, we can now propose the postulates that serve as the foundation for the new theory of Systemic Exergy Management (SYMΞX).

1. **|PS1| Thermoeconomic Foundation:** The fundamental ideas of thermoeconomics provide a comprehensive understanding of the utilization of resources within organizational contexts. This approach elucidates the physical principles that govern energy transitions and the economic consequences of resource utilization by integrating thermodynamics with economic analysis.
2. **|PS2| Systemic Sustainability:** The functioning of businesses is contingent upon the existence of intricate structures, which are characterized by the interconnectivity of various social, technical, economic, and environmental elements. Long-term success necessitates an acknowledgment of the impact of organizational actions across these various dimensions. Organizations are responsible for managing systemic sustainability, ensuring that their actions are aligned with the broader objectives and guided by thermoeconomic principles.
3. **|PS3| Resource Optimization:** In contrast to profit-centric approaches, organizations prioritize the optimization of resources throughout the value chain. By minimizing environmental impact and ensuring long-term viability, organizations foster sustainable resource utilization practices through the use of thermoeconomic analysis.
4. **|PS4| Exergy Analysis as a Core Tool:** An essential indicator of the potential utilization of resources in a work environment is its exergy, which serves as the foundation for effective resource management. By employing exergy analysis based on thermal economic principles, organizations can identify inefficiencies in resource consumption and prioritize measures that optimize resource value. This enables them to establish a sustainable trajectory through the design of more efficient resource management strategies.
5. **|PS5| Continuous Improvement and Innovation:** It is imperative for organizations to consistently enhance their environmental practices and resource efficiency, given the ever-evolving nature of sustainability concerns. Organizations that espouse an innovative culture engage in the investigation of novel technologies and procedures that prevent the

deterioration of resources and ensure continued advancement.

6. **|PS6| Holistic Decision-Making:** Organizations employ an all-encompassing approach to decision-making that extends beyond the conventional economic criteria. The incorporation of environmental and social considerations into decision-making frameworks ensures the formulation of ethical and sustainable decisions, informed by thermoeconomic principles that take into account the long-term consequences of decisions.
7. **|PS7| Cross-Functional Collaboration:** The existence of organizational silos presents a significant challenge in comprehending the utilization of resources and the sustainability initiatives that are in place across the entire organization. Organizations that facilitate cross-functional collaboration are better positioned to integrate sustainability principles into all business operations, fostering alignment and synergies toward shared sustainability objectives.
8. **|PS8| Data-Driven Management:** Despite the growing emphasis on data-driven decision-making, intuition remains an important aspect of organizational decision-making processes. The use of sustainability indicators based on thermoeconomic concepts can provide deeper insights, facilitating well-informed decision-making and efficient sustainability operations.
9. **|PS9| Systemic Transformation:** Organizations seek to accelerate systemic transformation because they recognize the cumulative impact of their activities. An organization may facilitate change beyond its own boundaries by advocating for sustainable practices across the supply chain and broader environment, thereby advancing sustainability as a whole.
10. **|P10| Long-Term Perspective:** In contrast to those who are shortsighted in their success, corporations adopt a long-term perspective. In light of the impact of organizational actions on future generations, and in alignment with thermoeconomic principles, organizations demonstrate a robust commitment to sustainable resource utilization, ensuring long-term sustainability and success.

These postulates explicitly situate thermoeconomics at the core of SYMΞX, within the context of organizational theories. By employing exergetic analysis, organizations can make well-informed decisions that optimize resource use, minimize impacts on the environment and society, and contribute to the achievement of long-term systemic sustainability. Based on the aforementioned postulates, a conceptual framework of Systemic Exergy Management (SYMΞX) was constructed using graph theory. Each postulate (PS 1 ÷ PS 10) is represented as a node, and the relationships between these postulates are represented as arcs connecting them (Fig. 2).

Relationships between postulates (arcs) are identified and described below:

- **PS1 → PS2 and PS3:** Thermoeconomic principles provide the foundation for both systemic sustainability (PS2) and resource optimization (PS3). They inform the understanding of environmental and economic implications (PS2) and guide the optimization of resource use (PS3).
- **PS2 → PS6:** Achieving systemic sustainability (PS2) necessitates holistic decision-making (PS6). This requires considering environmental and social consequences alongside economic factors.
- **PS3 → PS4:** Resource optimization (PS3) is facilitated by exergy analysis (PS4). By quantifying usable work potential, exergy analysis identifies inefficiencies in resource use, aiding optimization efforts.
- **PS4 → PS5:** Exergy analysis (PS4) informs continuous improvement and innovation (PS5). Understanding inefficiencies revealed by exergy analysis fuels the search for new technologies and processes that minimize resource degradation.
- **PS5 & PS7 → PS3:** Both continuous improvement (PS5) and interdepartmental collaboration (PS7) contribute to resource optimization (PS3). Innovation and collaboration lead to better resource management practices.
- **PS6 and PS8 → PS9:** Holistic decision-making (PS6) and data-driven management (PS8) are crucial for systemic transformation (PS9). By considering all factors and leveraging data, organizations can influence sustainable practices beyond their boundaries.
- **PS7 and PS8 → PS6:** Interdepartmental collaboration (PS7) and data-driven management (PS8) facilitate holistic decision-making (PS6). Collaboration ensures diverse perspectives are included, and data provides the foundation for informed choices.
- **PS2, PS3, PS5, PS6, PS8, and PS9 → PS10:** All these postulates (PS2, PS3, PS5, PS6, PS8, & PS9) contribute to a long-term perspective (PS10). By focusing on sustainability, resource optimization, and ethical decision-making, organizations ensure long-term viability.
- **PS10 → PS3:** A long-term perspective (PS10) reinforces resource optimization (PS3). By considering intergenerational impacts, organizations prioritize sustainable resource use practices.

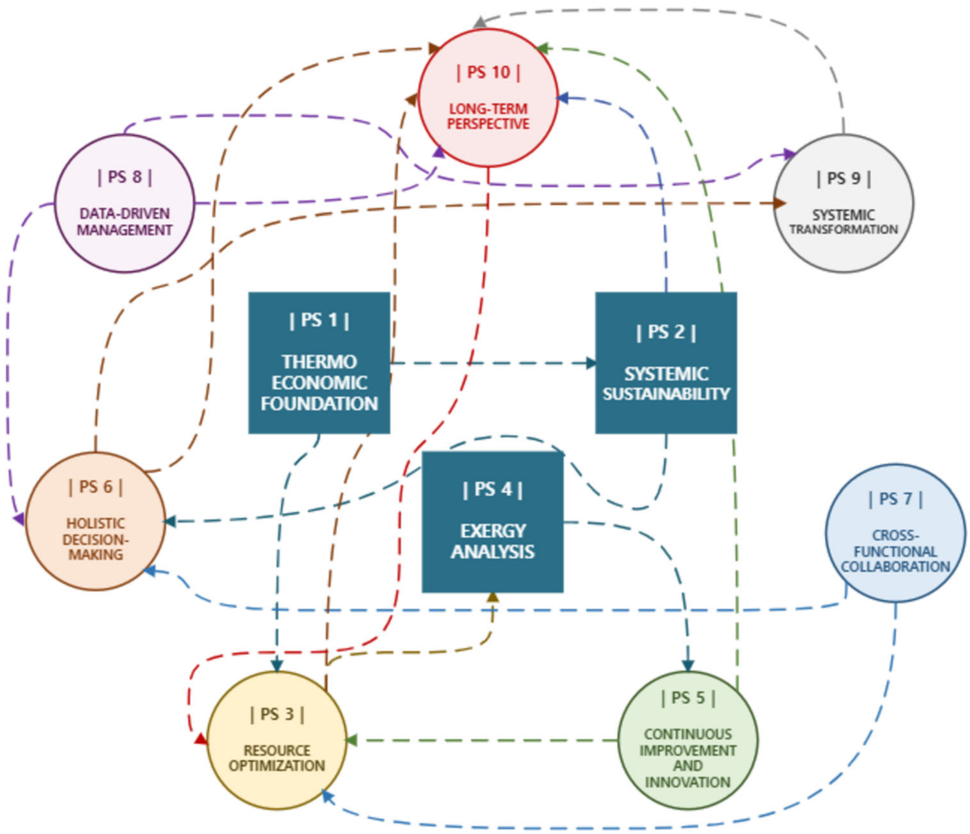


**Fig. 2** Conceptual Framework of Systemic Exergy Management (SYMΞX) based on Graph Theory

## Mathematical Framework

Figure 2 presents a graph-based conceptual framework that illustrates Systemic Exergy Management (SYMΞX) as a holistic theory with a robust foundation in thermoeconomics. The framework elucidates the interconnection between resource optimization, data-driven decision-making, cross-functional collaboration, and a long-term perspective. In light of the aforementioned postulates and relationships, a proposed description of the SYMΞX model is presented, employing a set of mathematical functions.

*Setup:*

- **Resources (R):** $R = \{r_1, r_2, \ldots, r_n\}$ (where $r_i$ represents a specific type of resource, e.g., raw material, energy source)
- **Processes (P):** $P = \{p_1, p_2, \ldots, p_m\}$ (where $p_i$ represents a specific process within the value chain, e.g., manufacturing, transportation)
- **Decisions (D):** $D = \{d_1, d_2, \ldots, d_n\}$ (where $d_i$ represents a specific decision with potential outcomes, e.g., investment in new technology)
- **Impacts (I):** *I = {I_env, I_social, I_economic, I_tech}* (where *I_env* represents environmental impact, *I_social* represents social impact, *I_economic* represents economic impact, and *I_tech* represents technological impact)
- **Metrics (M):** *M = {M_exergy, M_efficiency, M_cost, M_sustainability}* (where *M_exergy* represents exergy content, *M_efficiency* represents resource use efficiency, *M_cost* represents economic cost, and *M_sustainability* represents a composite systemic sustainability score)

Functions:

(i) Exergy analysis (EA): R → real number

$$EA(r\_i) = \Sigma\,(x\_j * exergy\_content(j, state(r\_i)))$$

where:

- *i* refers to the specific resource (*r_i*)
- *j* refers to each component within the resource (e.g., chemical elements in a material)
- *x_j* represents the mass or volume fraction of component j within *r_i*
- exergy_content(j, state(*r_i*)) is a pre-defined function that retrieves the exergy content per unit of component j based on its thermodynamic state (e.g., temperature, pressure) obtained from state(*r_i*))

  This function calculates the exergy content of a resource, representing its usable work potential.

(ii) Resource optimization (RO): P x R → R'.
This function identifies the optimized state of a resource for a specific process, minimizing exergy degradation.

$$RO(p\_j, r\_i) = minimize\,(\Delta EA(r\_i, p\_j))$$

where:

- *j* refers to the specific process (*p_j)*
- *i* refers to the specific resource (*r_i*)
- *ΔEA(r_i, p_j) = EA(r_i')—EA(r_i)* represents the change in exergy content after applying process *p_j* to resource *r_i,* resulting in the optimized resource state *r_i*')

(iii) Sustainability assessment (SA): D x I → Real Number
This function calculates a sustainability score for a decision, considering its environmental, social, economic, and technological impacts.

$$SA(d\_k, I\_l) = w\_l * f\_impact(d\_k, I\_l)$$

where:

- *k* refers to the specific decision (*d_k*)
- *l* refers to the specific impact category (*I_l*)—environmental, social, or economic
- *w_l* represents the weight assigned to impact category *l*, reflecting its relative importance (weights can be determined based on organizational priorities or industry standards)
- *f_impact(d_k, I_l)* is a pre-defined function that quantifies the impact of decision *d_k* on impact category *I_l*. This function can involve various metrics and calculations depending on the specific impact being assessed (e.g., life cycle assessment for environmental impact, social impact assessment metrics for social impact, and economic cost calculations)

(iv) Holistic decision-making (HDM): D x M -> D'
This function selects the most sustainable and cost-effective decision option by balancing economic cost and a composite sustainability score.

$$HDM(d\_k, M) = \arg\max\,(h(M\_\cos t(d\_k), M\_sustainability(d\_k)))$$

where:

- *k* refers to the specific decision (*d_k*)
- *M = {M_exergy, M_efficiency, M_cost, M_sustainability}* represents the set of relevant metrics
- *h represents a weighting function. Its purpose is to combine the economic cost (M_cost(d_k)) and the composite sustainability score (M_sustainability(d_k)) into a single value that can be used to compare different decision options (d_k).*
- *M_cost(d_k)* is a function that calculates the economic cost associated with decision *d_k*
- *M_sustainability(d_k) = Σ (w_l * SA(d_k, I_l))* is a composite sustainability score for decision *d_k*, calculated

(xxii) Integration of long-term perspective
The long-term perspective can be incorporated into the model by introducing discounting factors into the economic cost function (M_cost). Discounting future costs reflects the time value of money and prioritizes current investments with long-term benefits. The discount rate would need to be carefully chosen, balancing immediate financial considerations with long-term sustainability goals. The economic cost function (*M_cost)* could be modified as follows

$$M\_\cos t(d\_k) = immediate\_\cos t(d\_k) + \Sigma\,[f(discount\_rate, year\_i) * future\_\cos t(d\_k, year\_i)]$$

where:

- *immediate_cost(d_k)* represents the immediate economic cost of decision *d_k*. Represents the immediate economic cost of decision d_k (e.g., investment cost).
- *future_cost(d_k, year_i)* represents the economic cost of decision *d_k* in year *i*. This could include factors like resource depletion costs, maintenance costs of long-lived assets, or potential liabilities arising from environmental or social consequences.
- *f(discount_rate, year_i)* is a year-specific discounting factor based on the chosen discount rate, decreasing the weight of future costs as time goes on.

Relationships:

1. **RO(P, EA(R)) = R':** The optimized resource state (R') is achieved by applying the resource optimization function (RO) that utilizes exergy analysis (EA) to minimize exergy degradation for each process-resource combination.
2. **SA(D, I) influences HDM(D, M):** The sustainability assessment (SA) provides the sustainability score

(M_sustainability) as an input for the holistic decision-making function (HDM).

3. **HDM(D, M) considers PS1 (Thermoeconomic Foundation):** The function (h) used in HDM can incorporate a factor that reflects the economic value of the exergy content (M_exergy) derived from the resource analysis (EA), aligning with thermoeconomic principles.
4. **Long-term perspective (PS10) influences D:** Decision options (D) can be evaluated using the Economic Cost Function (M_cost) that consider intergenerational impacts, reflecting a long-term perspective.

Central to the SYMΞX model is the complex interplay between four key relationships; a more extensive discussion of these relationships is provided below.

1. **Resource optimization (RO) driven by exergy analysis (EA):** This relationship forms the foundation for efficient resource utilization. By leveraging exergy analysis (*EA*), which quantifies a resource's usable work potential, organizations can identify the most optimized state (*R'*) for each resource (*R*) within a specific process (*P*). Minimizing exergy degradation through *RO* ensures that resources are used effectively, reducing waste and maximizing their value throughout the value chain.
2. **Systemic sustainability assessment (SA) informing holistic decision-making (HDM):** This relationship ensures that environmental, social, economic, and technological considerations are woven into the fabric of decision-making. The sustainability assessment (*SA*) provides a holistic score (*M_sustainability*) that reflects the impact (*I*) of a decision (*D*) across all these crucial dimensions. This score, along with economic cost information (*M_cost*), feeds into the holistic decision-making function (*HDM*). *HDM* then prioritizes options that not only deliver economic benefits but also contribute to a sustainable future.
3. **Thermoeconomic foundation (PS1) shaping HDM:** This relationship ensures that economic considerations are intertwined with the laws of thermodynamics. The function (*h*) used in *HDM* can incorporate a factor that reflects the economic value of the exergy content (*M_exergy*) derived from the resource analysis (*EA*). This aligns with the fundamental principles of thermoeconomics, where resource utilization is optimized considering both its physical properties and economic viability.
4. **Long-term perspective (*PS10*) influencing decision-making (*D*):** This relationship ensures that decisions are not driven solely by short-term gains. The Long-Term Perspective principle (*PS10*) is reflected in the economic cost function (*M_cost)*. By considering intergenerational impacts and incorporating future costs (discounted based on a chosen rate), decision options (*D*) are evaluated with a long-term lens. This ensures that decisions made today contribute to a sustainable future for generations to come.

This model provides a mathematical framework for elucidating the fundamental concepts underlying SYMΞX. The model permits the quantitative examination of resource consumption, decision-making processes, and the implications of sustainability by employing functions and sets. It may be expanded by incorporating precise mathematical formulae or optimization algorithms for decision-making and resource management, tailored to specific use scenarios. SYMΞX proposes that decisions be made with a long-term vision in mind, taking into account economic viability, sustainability in its various forms, and resource efficiency. Organizations may utilize this interwoven web of ties to facilitate the decision-making process in a manner that is supportive of sustainability.

## Contribution to the Development of Management Theories

In comparison to conventional management theories, Systemic Exergy Management (SYMΞX) represents a pioneering approach to sustainability management. The primary objectives of conventional theories of productivity enhancement are the establishment of a hierarchical structure, the optimization of efficiency, and the standardization of norms (Borim-de-Souza et al., 2020). These concepts have been pivotal in shaping the discourse on organizational responsibility; however, they have also been subject to critique for their failure to adequately address the impact of organizational actions on society and the environment.

### Neoclassical Economics

Friedman's theory places emphasis on profit maximization and shareholder dominance, which can potentially encourage unsustainable behavior (Neck, 2022). SYMΞX challenges the aforementioned idea by promoting a global strategy that considers the interdependence of environmental, economic, technical, and social factors. This guarantees that businesses can maintain economic viability while operating within the ethical limits of the environment and society..

### Scientific Management

This is a seminal theory of management, first proposed by Frederick Taylor. The objective is to enhance efficiency by employing a scientific approach to the analysis of work processes (Hill & Van Buren, 2018). The approach emphasizes standardization, specialization and the division of labor, with the overarching objective of maximizing productivity and minimizing costs. In contrast to the principles of scientific management, SYMΞX places a greater emphasis on sustainability. While scientific management is primarily concerned with enhancing productivity, SYMΞX integrates the tenets of thermoeconomics and exergetic analysis to optimize the utilization of resources and minimize environmental impact. Additionally, SYMΞX underscores the significance of long-term resilience and ethical decision-making, which are not central tenets of scientific management.

### Administrative Management

The theory was developed by Henri Fayol and is centered on organizational structure and managerial functions (Pataki & Sagi, 2009). The five key managerial responsibilities are as follows: planning, controlling, commanding, coordinating, and organizing. The theory prioritizes a hierarchical structure and clear lines of authority with the objective of improving efficiency and fostering cooperation within the company. SYMΞX's incorporation of systemic principles of sustainability into management processes represents a more comprehensive approach than that of administrative management, which primarily focuses on coordination and organizational structure. The focus of administrative management is on the internal workings of the company, whereas SYMΞX considers the environmental and social aspects of the business. SYMΞX's management strategy is characterized by an emphasis on resource efficiency and system sustainability, which facilitates a more comprehensive approach to organizational management.

### Bureaucracy

In this theory, Weber directs attention to standardized procedures and hierarchical structures that may contribute to rigidity and impede adaptability to sustainability problems (Sager & Rosser, 2021). In contrast, SYMΞX fosters creativity and continuous improvement. Such an approach encourages the development of a culture of adaptation and flexibility, which enables businesses to continually enhance their systemic sustainability measures and resource efficiency.

### Human Relations Theory

The theory was initially proposed by Mayo, who developed it in response to the perceived shortcomings of scientific and administrative management theories (Bruce & Nyland, 2011). The theory underscores the pivotal role of social variables, worker happiness, and motivation. Furthermore, it encourages open communication, employee well-being, and participatory decision-making, underscoring the significance of interpersonal connections and the human aspect of business. Although human relations theory acknowledges the role of social elements in organizational efficiency, it does not address sustainability challenges in any specific manner. In contrast, SYMΞX acknowledges the interconnection between corporate actions and social and environmental consequences by integrating human and social capital into managerial strategies. SYMΞX endorses the principles of social resource efficiency, creativity, and teamwork, which are aligned with the objectives of sustainable management.

### Agency Theory

The primary focus of agency theory is the analysis of the relationship between principals and agents within an organizational context. This field of study also places significant emphasis on the mitigation of conflicts of interes (Eisenhardt, 1989). While agency theory places an emphasis on coordinating incentives and oversight procedures to reduce agency issues, SYMΞX places a higher priority on efficacy than efficiency. SYMΞX provides a more comprehensive perspective than the agency relationship by integrating concepts from thermoeconomics and exergetic analysis to optimize resource utilization and minimize environmental impact.

### Theory of Value

The primary objective of value theory (Pirgmaier, 2021) is the creation and capture of value, frequently through competitive advantage, innovation, and consumer value propositions. In contrast to the primacy of morality and sustainability in decision-making, value theory prioritizes the generation of value and the attainment of competitive advantage. Although both theories aim to enhance organizational performance, SYMΞX adopts a more comprehensive strategy by integrating principles of systemic sustainability into business models.

### Transaction Cost Theory

The costs associated with interactions between individuals or organizations are examined by transaction cost theory, along with the governance systems that may be employed to reduce these costs (Cuypers et al., 2021). The objective of transaction cost theory is to enhance efficiency and reduce costs by optimizing transactional procedures and governance frameworks. In contrast, SYMΞX places a greater emphasis on sustainability, pursuing optimization of resources, reduction of environmental impact, and a commitment to ethical decision-making and long-term resilience (De Marchi et al., 2023).

### Schumpeterian Theory

The Schumpeterian theory (Emami Langroodi, 2017) elucidates the role of entrepreneurship and innovation in propelling economic growth and organizational transformation. SYMΞX is predicated on the principles of resource efficiency and technical innovation. In contrast, Schumpeterian theory underscores the pivotal role of innovation and creative destruction in organizational dynamics. The two theories both emphasize innovation, but they do so in different ways. While Schumpeterian theory places greater emphasis on economic development, SYMΞX places a higher priority on systemic sustainability.

### Stakeholder Theory

In accordance with the tenets of stakeholder theory, it is incumbent upon companies to consider the interests of all stakeholders while making decisions (Mahajan et al., 2023). While SYMΞX espouses a life cycle strategy to engage stakeholders in the value chain, stakeholder theory places greater emphasis on stakeholder management and the broader social implications of organizational actions. Although both theories espouse a more comprehensive approach to organizational decision-making, SYMΞX places particular emphasis on systemic sustainability.

### Systemic Theory

The systemic theory of thought posits that organizations are complex systems that interact with their surroundings and are comprised of interdependent components that are themselves part of a larger systemic framework (Amagoh, 2016). SYMΞX shares the systemic school's recognition of the interconnection between organizational components and their broader environmental context. Although the systemic school provides a comprehensive framework for understanding organizational complexity, SYMΞX offers specific guidelines and tools for optimizing resource efficiency and minimizing environmental impact in organizational systems.

### Resource-Based View

In accordance with the tenets of the resource-based theory, the distintive and valued resources and capabilities of a corporation serve as the foundation for its competitive advantag (Davis & DeWitt, 2021). The resource-based theory is concerned with the utilization of an organization's internal resources and competencies, frequently through the consolidation of resources and the strategic allocation of assets, in order to gain a competitive advantage. SYMΞX, in contrast, places greater emphasis on the optimization and sustainability of resources, with a particular focus on resource efficiency and the reduction of environmental impact. While both theories acknowledge the relevance of organizational resources, resource-based theory places greater emphasis on competitive advantage, whereas SYMΞX prioritizes environmental sustainability.

To compare SYMΞX with classical organizational theories, we can visualize their key features and distinguishing characteristics in a comparative chart (Table 4).

The visual representation serves to elucidate the pioneering contributions of Systemic Exergy Management (SYMΞX) to the domain of sustainability management. To this end, it compares and contrasts the principles and approaches espoused by SYMΞX with those of classical organizational theories.

## Enhancing Organizational Flexibility Through SYMΞX

The conventional approach to organizational flexibility has been to adapt to changing consumer preferences and market circumstances. However, the imperative of addressing sustainability concerns necessitates a more comprehensive definition that incorporates environmental considerations. One useful framework for enhancing this adaptability is provided by Systemic Exergy Management (SYMΞX).

SYMΞX employs exergy analysis to identify opportunities for reducing resource utilization and waste generation, thereby reducing reliance on specific resources and enhancing flexibility in response to price fluctuations. Furthermore, SYMΞX advocates the selection of suppliers based on the exergy content of their products, thereby expanding the pool of suppliers and mitigating the associated risks of excessive dependence. Furthermore, SYMΞX facilitates process adaptation. It allows businesses to optimize energy inefficiencies in manufacturing processes, thereby enhancing their flexibility in responding to environmental regulations or fluctuating energy prices. Moreover, SYMΞX concepts have inspired the development of environmentally friendly technology with the objective of reducing exergy breakdown. This encourages

**Table 4** Visual representation comparing Systemic Exergy Management (SYMΞX) with classical organizational theories

| Theory | Key features | New contribution to sustainability management |
|---|---|---|
| Neoclassical economics | Emphasizes shareholder supremacy and profit maximization<br>Often leads to unsustainable practices | Challenges the notion of shareholder supremacy<br>Advocates for a holistic approach considering environmental, economic, technological, and social aspects |
| Scientific management | Focuses on efficiency through scientific analysis of work processes<br>Emphasizes standardization, specialization, and division of labor | Prioritizes sustainability over efficiency<br>Integrates principles of thermoeconomics and exergetic analysis for resource optimization |
| Administrative management | Focuses on organizational structure and management functions<br>Emphasizes hierarchical structure and clear lines of authority | Takes a broader perspective, integrating systemic sustainability principles into management practices<br>Considers external environmental and social impacts |
| Bureaucracy | Focuses on hierarchical structures and standardized processes<br>Can create rigidity, hindering adaptation to sustainability challenges | Promotes continuous improvement and innovation for flexibility and adaptability<br>Fosters a culture of systemic sustainability practices |
| Human relations | Emphasizes social factors, employee motivation, and satisfaction<br>Advocates participatory decision-making and open communication | Promote human and social capital into management practices<br>Promotes social resource efficiency, innovation, and collaboration |
| Agency theory | Focuses on mitigating conflicts of interest between principals and agents<br>Emphasizes aligning incentives and monitoring mechanisms | Prioritizes effectiveness over efficiency<br>Integrates principles from thermoeconomics and exergetic analysis |
| Theory of value | Centers on value creation and competitive advantage<br>Emphasizes innovation, customer value propositions, and organizational change | Prioritizes sustainability and ethical decision-making over competitive advantage<br>Integrates systemic sustainability principles into innovative business model |
| Transaction cost theory | Examines costs associated with transactions and governance mechanisms<br>Focuses on optimizing transactional processes to reduce costs | Prioritizes sustainability by minimizing environmental impact<br>Emphasizes long-term resilience and ethical decision-making |
| Schumpeterian theory | Emphasizes innovation and entrepreneurship as drivers of growth and change<br>Highlights creative destruction in organizational dynamics | Focuses on technological innovation and resource optimization<br>Prioritizes systemic sustainability over economic growth |
| Stakeholder theory | Suggests considering interests of all stakeholders in decision-making<br>Emphasizes stakeholder management and societal impact of decisions | Advocates for a broader perspective on organizational decision-making<br>Focuses on a life cycle approach to engage stakeholders in the value chain |
| Systemic theory | Views organizations as complex systems interacting with environments<br>Emphasizes interdependence of organizational elements and systemic context | Recognizes interconnectedness of organizational elements and environmental context<br>Offers specific principles and tools for sustainability practices |
| Resource-based view | Suggests competitive advantage based on unique and valuable resources<br>Focuses on leveraging internal resources for competitive advantage | Prioritizes systemic sustainability and resource optimization<br>Emphasizes effective use of resources and minimizing environmental impact |

creativity and facilitates the development of solutions to environmental challenges that will emerge in future. In addition to promoting operational and financial adaptability, SYMΞX also encourages stakeholder adaptability. Furthermore, it encourages the participation of stakeholders by integrating environmental concerns into core business operations. This open and cooperative approach enables businesses to adapt to evolving stakeholder expectations regarding environmental responsibility. Furthermore, SYMΞX facilitates the development of collaborative sustainability solutions through partnerships with key stakeholders, including suppliers, customers, and communities. This collaborative approach enhances a company's ability to navigate complex environmental challenges through collective action.

At last, SYMΞX transcends conventional resource management, thereby providing companies with greater general adaptability. The promotion of resource efficiency, process optimization, and stakeholder involvement enables SYMΞX to facilitate the adaptation and flourishing of businesses in the context of mounting environmental limitations. The long-term success and viability of an organization are contingent upon this enhanced flexibility.

## Discussion and Limitations

This study introduces the Systemic Exergy Management (SYMΞX) framework as a novel theoretical model that integrates thermoeconomic principles into business performance measurement. By employing exergy analysis, SYMΞX presents a comprehensive methodology for evaluating organizational performance that transcends the limitations of conventional financial metrics. It incorporates considerations of resource efficiency, environmental impact, and social responsibility, offering a more holistic and multifaceted approach to assessing organizational impact. The framework's capacity to align business decisions with both economic and environmental imperatives enhances organizational flexibility, enabling companies to more effectively adapt to resource scarcity and evolving environmental regulations. In this context, SYMΞX makes a significant contribution to the field of business scholarship, addressing shortcomings in performance measurement systems and promoting advances in sustainability practices.

The practical implications of SYMΞX are of considerable consequence. Organizations that adopt this framework are able to develop comprehensive Sustainability Performance Indicators (SPIs) that reflect not only financial outcomes, but also technological, environmental, and social dimensions. The integration of these indicators into performance measurement allows companies to demonstrate a deeper commitment to corporate responsibility, thereby improving stakeholder trust and reputation. The SYMΞX framework integrates exergy analysis into business performance measurement, providing a holistic perspective on organizational impact. Furthermore, it advances flexible systems management by aligning corporate decision-making with both financial and ecological imperatives, thereby enhancing organizational flexibility in response to resource scarcities and environmental regulations. SYMΞX plays a pivotal role in advancing flexible systems management by offering a framework that adapts to dynamic sustainability challenges through the lens of exergy efficiency. SYMΞX enhances organizational flexibility by aligning corporate decision-making with both financial and ecological imperatives, thereby enabling businesses to respond to resource scarcities and environmental regulations. Recent studies (Ma et al., 2022; A. Singh et al., 2023; S. Singh et al., 2021a, b) highlight the necessity for flexible, data-driven management models that integrate environmental considerations. SYMΞX addresses this gap through its multi-dimensional sustainability assessment.

While the SYMΞX framework offers several advantages, it is not without its own inherent limitations. Firstly, the framework's reliance on thermoeconomic principles, particularly exergy analysis, may prove challenging for industries lacking the requisite technical expertise in thermodynamics. The complexity of exergy calculations may impede the framework's adoption, particularly in sectors where environmental metrics are not yet fully integrated into decision-making processes. Moreover, SYMΞX has not yet been subjected to comprehensive empirical validation across a range of industrial sectors. The extent to which it can be applied in addressing sector-specific sustainability challenges remains an open question that requires further investigation. Moreover, while SYMΞX offers a robust theoretical framework, the translation of its principles into actionable, user-friendly decision-making tools for businesses may prove challenging. In light of the inherent complexities of thermoeconomics, it is imperative that future endeavors prioritize the accessibility of SYMΞX to a broader business community, including small- and medium-sized enterprises. It is imperative that researchers investigate methods to streamline exergy analysis for practical applications. This could be achieved by developing decision-support software or tools that integrate SYMΞX with existing management systems. Furthermore, additional research is required to examine the integration of SYMΞX with global reporting frameworks, such as the European Sustainability Reporting Standards (ESRS), and to ascertain how it can enhance existing sustainability management systems.

## Conclusion

This study introduces the Systemic Exergy Management (SYMΞX) framework, which represents a pioneering model that integrates thermoeconomic principles into business performance measurement. SYMΞX offers a comprehensive approach to sustainability, encompassing not only financial outcomes but also resource efficiency, environmental impact, and social responsibility. By employing exergy analysis, this framework facilitates a more profound comprehension of an organization's comprehensive impact, thereby enabling more judicious decision-making that strikes a balance between profitability and sustainability objectives. The SYMΞX framework contributes to the advancement of business science by filling gaps in traditional performance assessment methods and providing a flexible, adaptable model for addressing contemporary sustainability challenges. The framework assists organizations in aligning their operations with both financial and environmental objectives, thereby enhancing organizational flexibility and resilience in response to environmental regulations and resource constraints. In future, research should focus on empirically validating SYMΞX across different industries to test its versatility in different contexts. Moreover, the creation of user-friendly instruments for incorporating SYMΞX into routine business operations will be vital to enhancing its practical implementation. The path to systemic sustainability necessitates the implementation of frameworks such as SYMΞX, which serve to direct organizations toward the adoption of more ethical and resource-efficient practices. Adoption of the SYMΞX framework enables companies to make a significant contribution to a sustainable future while maintaining competitiveness and financial success.

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## Key Questions

1. What are the limitations of traditional economic models in considering resource efficiency and environmental impact in corporate performance assessments?
2. How can the integration of thermoeconomic principles bridge the existing gaps in corporate performance evaluation models?
3. What are the theoretical and practical challenges in applying exergy analysis as a measure of resource efficiency in organizations?
4. How can the SYMΞX framework contribute to a more comprehensive understanding of systemic sustainability compared to current approaches?
5. What are the implications of adopting the SYMΞX framework for organizational flexibility and adaptation to evolving economic and environmental conditions?
6. What gaps exist in the current literature regarding the use of exergy as a sustainability performance indicator, and how can the SYMΞX framework address them?

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